Personalized Learning Recommendation System Documentation

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Roll Number: BWU/BCA/24/076

Subject: Machine Learning (Subject Code: BCA57205)

Programming Language: Python | **Framework:** Streamlit | **Algorithm:** Random Forest Classifier

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1. Introduction

1.1 What is this project?

The **Personalized Learning Recommendation System** is a Machine Learning project. It helps a student find the right study material. The student gives some information about themselves, such as their marks, what they like to study, how they like to learn, and what they have already finished. The system then looks at this information and suggests **five learning resources** that suit that student best.

Think of it like a smart study friend. If you tell a friend, "I scored 70 marks, I like Data Science, I learn best by watching videos, and I have already finished Python basics," a good friend would suggest what to study next. This project does the same thing, but using a computer and a trained Machine Learning model.

1.2 Why do we need it?

Today there are thousands of videos, notes, websites and practice sheets available online. This is good, but it also creates confusion. A student often does not know:

- Which resource is right for me?
- Is this too hard or too easy for me?
- What should I study next?
- Have I already studied this?

A normal learning website shows the same list of resources to every student. But every student is different. One student may be strong in mathematics and weak in programming. Another may like videos while someone else prefers practice questions. A system that treats all of them the same will not help everyone.

1.3 What does this project do?

This project solves the problem by creating **personal recommendations**. It does not give one common list to everyone. It reads the profile of each student and creates a special list for that student only.

1.4 Short summary

Question	Simple Answer
What is the project?	A system that recommends study resources to students
Who is it for?	Students who want to find suitable learning material
What does the student give?	Score, interests, learning style, completed topics
What does the system give?	Five best learning resources with suitability scores
Which algorithm is used?	Random Forest Classifier
How does the student use it?	Through a Streamlit web application

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2. Project at a Glance

The table below gives all the main details of the project in one place.

Particulars	Details
Project Title	Personalized Learning Recommendation System
Developed By	Priyasmita Karmakar
Project Domain	Machine Learning
Subject	Machine Learning
Subject Code	BCA57205
Programming Language	Python
Framework	Streamlit
ML Algorithm	Random Forest Classifier
Dataset	Student Learning Interactions
Recommendation Type	Personalized Resource Recommendation
Main Objective	To develop an intelligent recommendation system that analyzes student profiles and recommends the five most suitable learning resources

2.1 Explanation of each detail

Project Title. The name of the project is Personalized Learning Recommendation System. "Personalized" means made for one person. "Learning" means studying. "Recommendation" means suggesting. So the full name means *a system that suggests study material made for each person*.

Project Domain. The domain is Machine Learning. This means the project uses a computer program that learns from data instead of following only fixed rules.

Programming Language. Python is used because it is simple to read and has many ready-made libraries for data and Machine Learning.

Framework. Streamlit is used to turn the Python code into an interactive web page, so a student can use the project without writing any code.

ML Algorithm. Random Forest Classifier is the Machine Learning method used to judge how suitable a resource is for a student.

Dataset. The project uses a dataset called Student Learning Interactions, along with a dataset of learning resources. Both are explained in detail later.

Recommendation Type. The system gives *personalized resource recommendations*. That means each student receives suggestions based on their own profile.

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3. Problem Statement

3.1 The main problem

Every student has different learning abilities, interests and preferences. Traditional educational platforms often provide similar learning materials to all students, regardless of their individual requirements.

3.2 Problems this creates

This creates several problems. Each one is explained below in simple words.

Problem 1: Resources may be too difficult or too easy. If a beginner is given an advanced resource, they may feel lost and lose interest. If a strong student is given a very easy resource, they waste their time. In both cases learning is not effective.

Problem 2: Finding relevant material takes a lot of time. Students often spend a long time searching on the internet, opening many pages and checking which one is useful. This time could have been used for actual studying.

Problem 3: Students do not know what to study next. After finishing one topic, many students are unsure about the next step. Without guidance, they may jump to a topic that is too hard or repeat things they already know.

Problem 4: Learning preferences are ignored. Some people learn best by watching videos. Others like solving practice questions. If the platform ignores this, the student may not enjoy or understand the material as well.

Problem 5: Completed topics may be recommended again. If a student has already finished a topic, it is not helpful to suggest it again and again. It wastes time and feels boring.

3.3 Our solution

Develop a personalized recommendation system that evaluates student information and suggests suitable educational resources.

3.4 Problem and solution together

Problem	How this project helps
Resources too hard or too easy	Recommendations consider the student's academic score and resource difficulty
Searching takes time	The system gives a ready list of five resources
Not sure what to study next	The system suggests topics based on interests and completed topics
Learning style ignored	The student chooses a preferred learning style
Completed topics repeated	The student marks completed topics so they can be taken into account

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4. Aim, Objectives and Scope

4.1 Aim

The aim of this project is to build an intelligent recommendation system that studies a student's profile and recommends the five most suitable learning resources.

4.2 Objectives

The objectives are the smaller goals that help reach the main aim.

1. To collect and use a dataset of student learning interactions.
2. To prepare and understand the data using Exploratory Data Analysis (EDA).
3. To train a Random Forest Classifier that can evaluate learning suitability.
4. To test the model on separate test data and measure its performance.
5. To use a dataset of learning resources containing topics, types, difficulty and links.
6. To rank resources by their suitability scores and select the top five.
7. To build an easy Streamlit application where students can enter their profile and see recommendations.
8. To save the trained model so that it can be reused without training again.

4.3 Scope

The scope tells us what the project covers.

- It works with student profile information: score, interests, learning style and completed topics.
- It recommends from a collection of **104 learning resources**.
- It shows **five** recommendations at a time.
- It runs as an interactive Streamlit application.
- It is a learning and demonstration project built for the Machine Learning subject.

4.4 What is outside the scope

It is important to be honest about what the project does not do.

- It does not prove that students' marks will improve.
- It has not been tested on a large number of real students.
- It is not yet connected to a real online learning platform.

These points are discussed again in the Limitations section.

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5. Proposed Solution

5.1 Overview

The proposed solution is a Machine Learning-based application. It has three main parts working together:

Part	What it does
Data Analysis	Understands the student learning information

Part	What it does
Machine Learning	Evaluates how suitable a resource is for a student
Recommendation System	Ranks resources and suggests the best five

5.2 How the three parts work together

First, data analysis helps us understand the learning data. Next, the Machine Learning model is trained on that data so that it can judge suitability. Finally, the recommendation system uses the model to score the available resources and shows the top five to the student.

5.3 A simple real-life comparison

Imagine a library with 104 books. A normal library notice board lists all books for everyone. But our system is like a librarian who asks you a few questions:

- How well are you doing in your studies?
- What subject do you enjoy?
- Do you like reading, watching or practising?
- Which books have you already finished?

Then the librarian picks the five best books for you. That is what the Personalized Learning Recommendation System does.

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6. How the Project Works (Step by Step)

The application follows a clear workflow of five steps.

Step 1: Student Profile Input

The student provides the following information:

- **Academic score (0 to 100).** This shows how the student is performing in studies.
- **Learning interests.** The subjects or areas the student likes, for example Data Science.
- **Preferred learning style.** How the student likes to learn, for example by watching videos or by practising.
- **Previously completed topics.** The topics the student has already finished.

Step 2: Data Processing

The application takes the student's information and prepares it so that the recommendation system can use it. Computers work best with clean and organised data, so this step arranges the input in the form the model needs.

Step 3: Machine Learning Model

The trained Random Forest model evaluates learning suitability. In simple words, the model checks how well a learning resource fits the student's profile.

Step 4: Recommendation Generation

The system looks at the available educational resources and ranks them according to suitability. The resource with the highest suitability comes first.

Step 5: Final Results

The application displays the **five recommended learning resources** along with their suitability scores, so the student can choose and start learning.

6.1 Workflow in one line

Student Profile Input → Data Processing → Random Forest Model → Ranking of Resources → Top 5 Recommendations

6.2 A worked example

Let us take an imaginary student to understand the flow.

Input	Example Value
Academic score	A number between 0 and 100
Interest	Data Science
Learning style	Prefers a certain style such as video or practice
Completed topics	Topics the student has already finished

The application processes this information. The model checks the resources. The system ranks them. At the end, the student sees five resources, for example Statistics, Data Cleaning, EDA, Pandas and NumPy, each with a suitability percentage. A complete recorded example is given in Section 11.

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7. Dataset Details

A dataset is a collection of data stored in an organised form, usually a table with rows and columns. This project uses two main datasets.

7.1 Dataset A: Student Learning Interactions

- It contains **2,500 student interaction records**.
- It is used for **training and evaluating** the suitability model.
- It provides the learning-interaction information that the model needs.
- The file is stored as `student_learning_interactions.csv` inside the `data` folder.

What is an "interaction"? An interaction means a student's connection with a learning resource. For example, a student looking at a resource and the result of that connection is one interaction record. Having 2,500 such records gives the model enough examples to learn patterns.

How the 2,500 records are used. The data is divided so that the model learns from one part and is tested on a separate part. The model was tested on **500 test records**, which were kept separate from training. This is important because testing on data the model has never seen shows whether it really learned something useful.

Important note. The training dataset is **synthetic**. Synthetic means it was created artificially for the project and was not collected from real students. This is mentioned again in the Limitations section.

7.2 Dataset B: Learning Resources

- It contains **104 educational resources**.
- It is the list from which the system picks its recommendations.
- The file is stored as `learning_resources.csv` inside the `data` folder.

The columns of this dataset are shown below.

Column	Description
resource_id	Unique resource identification
category	Learning category
topic	Educational topic
resource_type	Video, practice, etc.
difficulty	Difficulty level

Column	Description
estimated_minutes	Estimated completion time
description	Resource description
url	Learning resource link

7.3 Meaning of each column in simple words

resource_id. Every resource has its own special number or code. Just like every student has a roll number, every resource has a resource_id. This helps the system tell resources apart.

category. This is the big group the resource belongs to, for example Data Science. It is like a section of a library.

topic. This is the specific subject of the resource, for example Statistics, Pandas or NumPy. If category is the shelf, topic is the book.

resource_type. This tells how the resource teaches. It can be a Video, a Practice resource, and so on. This helps match the student's preferred learning style.

difficulty. This tells how hard the resource is. It helps avoid giving a very hard resource to a beginner or a very easy one to a strong student.

estimated_minutes. This is the approximate time needed to finish the resource. It helps a student plan their time.

description. A short explanation of what the resource contains.

url. The web link where the student can open the resource.

7.4 Why two datasets?

The two datasets have two different jobs.

Dataset	Job
Student Learning Interactions (2,500 records)	Teaches the model how to judge suitability
Learning Resources (104 resources)	Provides the actual resources to recommend

You can think of the first as the *experience* from which the model learns, and the second as the *shop* from which the recommendations are chosen.

7.5 Data dictionary

The project also includes a file called `DATA_DICTIONARY.md` in the data folder. A data dictionary is a small guide that explains what each column in the data means. It is useful for anyone who wants to understand or reuse the data later.

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8. Machine Learning Basics Used in This Project

Before studying the algorithm, it helps to understand a few basic ideas in very simple words.

8.1 What is Machine Learning?

Machine Learning (ML) is a way of teaching computers to learn from data. In normal programming, a human writes exact rules. In Machine Learning, we give the computer many examples, and it finds the rules by itself.

Example: Instead of writing rules like "if marks are above 80 then recommend advanced topics", we show the computer many examples of students and outcomes. The computer learns the patterns and can then make decisions for new students.

8.2 What is Supervised Learning?

Supervised learning is a type of Machine Learning where the data comes with correct answers, called **labels**. The model learns by comparing its guesses with the correct answers, like a student learning with an answer key.

Random Forest, which is used in this project, is a **supervised** learning algorithm.

8.3 What is Classification?

Classification means putting things into categories. For example, deciding whether an email is spam or not spam is classification.

In this project, the Random Forest **Classifier** is used to evaluate learning **suitability**. In simple words, it helps decide whether a resource is suitable for a student.

8.4 Training and Testing

Term	Meaning
Training	The model learns patterns from the training data

Term	Meaning
Testing	The trained model is checked on separate data it has not seen before
Test records	The 500 records used for checking the model

Keeping test data separate is like giving students a surprise exam with new questions instead of the same questions they practised. It shows real understanding.

8.5 What is a Decision Tree?

A decision tree is a simple model that makes decisions by asking a series of questions, like a flowchart. For example: "Is the score above a certain value? If yes, go this way; if no, go that way." At the end of the path, it gives an answer.

One decision tree alone can make mistakes or depend too much on small details of the data. That is why we use a *forest* of many trees, explained in the next section.

8.6 What is a Recommendation System?

A recommendation system is a program that suggests items to a person. We see them in daily life: video apps suggest videos, shopping apps suggest products, and music apps suggest songs. In this project, the items being suggested are **learning resources**.

8.7 Two different ideas: classification and recommendation

It is very important to understand that this project has two connected but different parts:

1. **The classification model**, which is evaluated using accuracy, precision, recall and F1-score.
2. **The recommendation engine**, which gives ranked resources with suitability scores.

These two should never be confused. This is explained further in Sections 10 and 21.

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9. The Random Forest Classifier

9.1 What is Random Forest?

Random Forest Classifier is the Machine Learning algorithm used in this project. It is a supervised machine learning algorithm that combines multiple decision trees to make predictions.

The name is easy to understand. A *forest* is made of many *trees*. In the same way, a Random Forest is made of many decision trees. Each tree gives its own opinion, and the forest combines all the opinions to reach a final result.

9.2 A simple example to understand

Imagine you want to decide which movie to watch. If you ask only one friend, that friend may have a strange taste and give you a bad suggestion. But if you ask ten friends and follow what most of them say, your decision will be better. Random Forest works in the same way. Many trees give their answers and the combined answer is more reliable than the answer of one single tree.

9.3 How it works, step by step

1. The training dataset is provided to the algorithm.
2. Multiple decision trees are constructed.
3. Each tree learns patterns from the training data.
4. The model combines the outputs of these trees.
5. The resulting model is used to evaluate learning suitability.

Let us understand each step.

Step 1: The training dataset is given. We give the algorithm the Student Learning Interactions data, which has 2,500 records. This is the material the model learns from.

Step 2: Many decision trees are built. The algorithm does not build just one tree. It builds many trees. Each tree is slightly different from the others.

Step 3: Each tree learns patterns. Every tree studies the training data and learns its own set of rules. One tree may notice one pattern, and another tree may notice a different pattern.

Step 4: The outputs are combined. When we need a prediction, every tree gives its answer. The forest combines these answers into one final result. Because many trees take part, one tree's mistake is less likely to spoil the final answer.

Step 5: The model is used for suitability. The final trained model is then used to evaluate how suitable a learning resource is for a student.

9.4 Why was Random Forest chosen?

There are several reasons why Random Forest is a good choice for this project.

Reason	Meaning in simple words
Handles different types of features	Student data has numbers (like score) and categories (like learning style). Random Forest can work with both kinds of information
Can learn complex relationships	Student suitability does not depend on only one thing. The model can learn how several things work together
Reduces dependence on a single decision tree	Because many trees vote, one weak tree does not control the result
Provides a useful classification baseline	It is a strong, reliable starting method that gives a good first result to compare with other methods later
Works well for structured datasets	Our data is in neat rows and columns (structured), which suits Random Forest

9.5 Strengths and weaknesses in simple words

Strengths:

- It is generally easy to use and gives good results without too much tuning.
- It does not depend on one tree, so it is more stable.
- It works well with table-style data.

Weaknesses to be aware of:

- It is harder to explain than a single decision tree, because many trees are involved.
- It needs good data. If the training data is poor or artificial, the model's learning will also be limited.

The second weakness connects to our project, because our training data is synthetic. This is discussed in Section 18.

9.6 Role of Random Forest in this project

In this project, Random Forest is used to **evaluate learning suitability**. It is a *classification* model. It is not itself the whole recommendation system. The complete recommendation system uses the model's evaluation, together with the list of learning resources, to rank and show the best five.

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10. Model Training and Evaluation

10.1 What was done?

The model was trained using the training data and then **evaluated using a separate test dataset**. The test dataset had **500 records**. Because these records were not used for learning, they give an honest idea of how the model performs on new data.

10.2 The results

Evaluation Metric	Result
Accuracy	86.8%
Precision	94.06%
Recall	90.62%
F1-Score	92.31%
Test Records	500

10.3 Understanding the results in simple words

The four metrics measure different things. Let us understand each one.

Accuracy: 86.8%

Accuracy measures the proportion of correct predictions among all test predictions.

In simple words: out of all the test records, how many did the model get right? An accuracy of 86.8% means that, on the 500 test records, the model's predictions were correct about 86.8 times out of every 100.

Example: If a teacher checks 100 answers and 87 are correct, the accuracy is about 87%.

Precision: 94.06%

Precision measures how many positive predictions were correct.

In simple words: when the model says "this is suitable", how often is it really right? A precision of 94.06% means that when the model predicted a positive result, it was correct about 94 times out of 100.

Why it matters here: High precision means the model does not wrongly call unsuitable things suitable very often.

Recall: 90.62%

Recall measures how many actual positive cases were correctly identified.

In simple words: out of all the cases that really were positive, how many did the model find? A recall of 90.62% means the model found about 90.6 out of every 100 real positive cases and missed the rest.

Example: If there are 100 genuinely suitable matches and the model finds 91 of them, the recall is 91%.

F1-Score: 92.31%

F1-Score provides a combined measure of precision and recall.

In simple words: precision and recall are two different measures, and sometimes one is high while the other is low. F1-Score mixes both into one number. A high F1-Score means the model is good at both: being correct when it says positive, and finding most of the real positives.

10.4 A quick comparison table

Metric	Question it answers	Result
Accuracy	How many predictions were correct overall?	86.8%
Precision	When the model says positive, how often is it right?	94.06%
Recall	Of all real positives, how many were found?	90.62%
F1-Score	How good is the balance between precision and recall?	92.31%

10.5 The Confusion Matrix

The project saves a file called `confusion_matrix.csv`. A confusion matrix is a small table that shows where the model was right and where it made mistakes. It normally has four parts:

Part	Meaning
True Positive	The model said positive, and it really was positive
True Negative	The model said negative, and it really was negative
False Positive	The model said positive, but it was actually negative

Part	Meaning
False Negative	The model said negative, but it was actually positive

Accuracy, precision, recall and F1-Score are all calculated from these four parts. The project also saves a `classification_report.txt` file, which lists these metrics in a report format, and an `evaluation_metrics.json` file, which stores the final metric values.

10.6 Very important warning

> **These are classification metrics, not recommendation accuracy or proof of improved student performance.**

This sentence should be remembered carefully. The numbers above tell us how well the Random Forest model performed on the classification task using test data. They do **not** tell us:

- how accurate the final list of five recommendations is, or
- that students who use the system will get better marks.

Proving those two things would need testing with real students over a period of time, which has not been done yet.

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11. The Recommendation System

This is the main functionality of the project.

11.1 What does it do?

The system generates **five recommendations** according to the student's profile. It looks at the available learning resources, uses the model to judge suitability, ranks the resources, and shows the best five.

11.2 Recorded example

Consider a student who is interested in **Data Science**. The recorded example recommendations were:

Rank	Recommended Topic	Resource Type	Suitability
1	Statistics	Practice	87.3%
2	Data Cleaning	Practice	85.0%
3	EDA	Video	84.5%
4	Pandas	Video	84.1%
5	NumPy	Practice	82.3%

The recommendations are ranked according to their suitability scores. The highest score is shown first.

11.3 How to read this table

- **Rank 1 is Statistics (Practice) at 87.3%.** This is the resource the system found most suitable for this student.
- **Data Cleaning (Practice) is second at 85.0%.** It is slightly less suitable than Statistics, but still a strong match.
- **EDA (Video) and Pandas (Video) are third and fourth.** These are video resources, so the list mixes different learning formats.
- **NumPy (Practice) is fifth at 82.3%.** It is the last in the top five.

Notice that the list contains both Practice and Video resources. The system recommends **multiple learning formats**, not just one.

11.4 What information does each recommendation show?

The system also provides resource information such as:

Information	Why it is useful
Topic name	Tells the student what they will learn
Resource category	Shows which learning area it belongs to
Difficulty level	Helps the student know if it is right for their level
Estimated completion time	Helps the student plan their time
Resource description	Gives a short summary of the content
Learning URL	Gives the direct link to start learning
Suitability score	Shows how well the resource fits the student

11.5 What is a suitability score?

A suitability score is a percentage that tells how well a resource fits a particular student. A higher percentage means a better fit. It is used only to **rank** the resources. The top five by score are shown to the student.

11.6 Suitability score is NOT model accuracy

This is the most important point for your viva.

Term	What it is
Suitability score (for example 87.3%)	A score given to one recommended resource for one student
Model accuracy (86.8%)	How often the classification model's predictions were correct on 500 test records

These are different things measured in different ways. Never say "87.3% accuracy" for a recommendation. Say "87.3% suitability score".

11.7 Saving the latest recommendations

The latest recommendations are saved in a file called `outputs/last_recommendations.csv`. This allows the results to be checked or reused later without running the whole process again.

11.8 The role of completed topics

One of the problems in Section 3 was that completed topics may be recommended again. The student provides their previously completed topics as part of the profile. This information is part of the student profile used by the system, so that the recommendations can take the student's progress into account.

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12. Application Interface (Streamlit)

12.1 What is Streamlit?

Streamlit is a Python framework that turns Python code into an interactive web application. The developer does not need to write complex web code (like HTML and JavaScript). A few lines of Python can create buttons, sliders, drop-down menus and result displays.

12.2 Why Streamlit?

The project uses Streamlit to provide an interactive application. The interface allows students to interact with the recommendation system **without manually running Python scripts**. A student only needs to open the page, fill in the details and click a button.

12.3 The two functional areas

The application has two important functional areas.

A. Student Profile Section

In this section the student can:

- Enter their academic score.
- Select interests.
- Choose their learning style.
- Select completed topics.
- Generate recommendations.

B. Recommendation Results Section

In this section the application displays:

- Five recommended learning resources.
- Suitability percentages.
- Learning categories.

- Resource types.
- Difficulty information.
- Estimated time.
- Resource links.

12.4 How a student uses the application

1. Open the application in the browser.
2. Enter the academic score (0 to 100).
3. Select one or more learning interests.
4. Choose the preferred learning style.
5. Select the topics already completed.
6. Click the button to generate recommendations.
7. Read the five recommendations with suitability scores.
8. Click a resource link to start learning.

12.5 Input and output summary

Section	Inputs / Outputs
Student Profile Section (Input)	Academic score, interests, learning style, completed topics, a button to generate
Recommendation Results Section (Output)	Five resources, suitability %, category, type, difficulty, time, link

12.6 Main file of the application

The application code is in the file `app.py` inside the `src` folder. It connects the interface with the recommendation logic written in `recommender.py`, and it loads the saved model so that the model does not have to be trained again each time.

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13. Technologies and Libraries

13.1 Summary table

Technology	Purpose
Python	Main programming language

Technology	Purpose
Streamlit	Interactive web application
Pandas	Data manipulation
NumPy	Numerical operations
Scikit-learn	Machine learning
Random Forest	Classification algorithm
Joblib	Saving and loading trained models
Matplotlib	Data visualization

13.2 Explanation in simple words

Python. Python is the main programming language of the project. It is popular for Machine Learning because its code is easy to read and it has many helpful libraries.

Streamlit. Streamlit builds the interactive web application, so the project can be used by clicking and selecting rather than by writing code.

Pandas. Pandas is a library for working with tables of data. It is used to read the CSV files, look at the data, clean it and prepare it. Think of it as Excel inside Python.

NumPy. NumPy is a library for fast numerical calculations. It works with numbers and arrays and supports many data operations.

Scikit-learn. Scikit-learn is the Machine Learning library. It provides ready-made algorithms, including the Random Forest Classifier, and tools to measure performance.

Random Forest. This is the classification algorithm used to evaluate learning suitability. It is provided through Scikit-learn.

Joblib. Joblib is used to save the trained model into a file and load it again later. This saves time because training is done only once.

Matplotlib. Matplotlib is used for data visualization. It draws charts and graphs, which help in understanding the data during EDA.

13.3 How the technologies fit together

Stage	Technologies used
Reading and preparing data	Pandas, NumPy
Exploring data with charts	Pandas, Matplotlib

Stage	Technologies used
Training the model	Scikit-learn (Random Forest)
Saving the model	Joblib
Showing the application	Streamlit
Overall programming	Python

The file `requirements.txt` in the `src` folder lists the libraries that need to be installed to run the project.

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14. Project Folder Structure

A well-organised folder makes a project easy to understand and maintain. The project structure was organised approximately as follows.

```
Personalized Learning Recommendation System/
│
├── Personalized_Learning_Recommendation_System/
│   │
│   ├── data/
│   │   ├── student_learning_interactions.csv
│   │   ├── learning_resources.csv
│   │   └── DATA_DICTIONARY.md
│   │
│   ├── src/
│   │   ├── app.py
│   │   ├── recommender.py
│   │   ├── eda.py
│   │   └── requirements.txt
│   │
│   ├── outputs/
│   │   ├── suitability_model.joblib
│   │   ├── evaluation_metrics.json
│   │   ├── classification_report.txt
│   │   ├── confusion_matrix.csv
│   │   ├── last_recommendations.csv
│   │   └── EDA visualizations
│   │
│   ├── reports/
```

```
|   |   └── project_report_template.md
|   |
|   ├── README.md
|   └── Documentation
```

14.1 Folder-by-folder explanation

The data folder

File	What it contains
student_learning_interactions.csv	The 2,500 student interaction records used for training and testing
learning_resources.csv	The 104 educational resources used for recommendations
DATA_DICTIONARY.md	An explanation of what each data column means

The src folder (source code)

File	What it does
app.py	The Streamlit application that students use
recommender.py	The recommendation logic that ranks resources and picks the top five
eda.py	The module that explores the data and creates visualizations
requirements.txt	The list of libraries needed to run the project

The outputs folder (results)

File	What it contains
suitability_model.joblib	The saved trained model
evaluation_metrics.json	The saved performance numbers
classification_report.txt	The detailed classification report
confusion_matrix.csv	The table of correct and incorrect predictions
last_recommendations.csv	The most recent five recommendations

File	What it contains
EDA visualizations	Charts and graphs created during data analysis

The reports folder

The reports folder contains `project_report_template.md`, a template that helps in writing the project report.

Top-level files

- **README.md** gives a short introduction and instructions about the project.
- **Documentation** is the project documentation file.

14.2 Why a good structure matters

- Data, code, results and reports are kept separate, so nothing gets mixed up.
- Anyone opening the project can quickly find what they need.
- It is easier to fix problems and add new features later.

---

15. Data Analysis and Visualization (EDA)

15.1 What is EDA?

EDA stands for **Exploratory Data Analysis**. It means looking at the data carefully, using numbers and charts, before building the model. It is like reading a book's contents page before reading the whole book: you understand what you are dealing with.

15.2 Why is EDA needed?

EDA helps understand the available learning-interaction data before building the model. If we skip EDA, we may build a model on data we do not understand and miss problems such as strange values or uneven data.

15.3 The eda.py module

The `eda.py` module is responsible for generating visualizations and exploring the dataset. The charts it creates are saved in the `outputs` folder as EDA visualizations.

15.4 What we try to understand

The purpose is to understand:

Purpose	Simple meaning
Dataset characteristics	What does the data look like? How many records and columns are there?
Learning interaction patterns	How do students interact with learning resources?
Distribution of available information	How are the values spread? Are there many of one kind and few of another?
Relationships between relevant features	Does one feature change along with another?

15.5 Common types of charts used in EDA

The project uses Matplotlib for visualization. In general, EDA uses charts such as the following to understand data.

Chart type	What it helps to see
Bar chart	Comparison between categories, such as how many resources are in each category
Histogram	How numbers such as scores are spread across ranges
Pie chart	The share of each part in a whole
Scatter plot	The relationship between two number columns

The exact charts produced are saved with the project outputs.

15.3 Benefits of doing EDA

- It builds understanding of the data before modelling.
- It helps find unusual patterns or problems early.
- It makes the project report more clear, because charts explain things better than only numbers.

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16. Model Saving with Joblib

16.1 Why do we save the model?

Training a model takes time and effort. If we had to train the model again every time someone opens the application, it would be slow and wasteful. So, after training once, we **save** the model in a file. Later, the application simply **loads** the saved model and uses it immediately.

Think of it like cooking. You cook a big pot of food once and store it. Whenever someone is hungry, you just heat and serve, instead of cooking from the beginning again.

16.2 The saved model file

The trained model was saved using Joblib. The file is:

```
outputs/suitability_model.joblib
```

This allows the application to load the trained model without training it from scratch every time.

16.3 Other saved outputs

Other evaluation outputs include:

File	Purpose
evaluation_metrics.json	Stores the final performance numbers
classification_report.txt	Stores the detailed classification report
confusion_matrix.csv	Stores the confusion matrix

Your latest recommendations are also saved in:

```
outputs/last_recommendations.csv
```

16.4 Summary of the saving process

Stage	What happens
Train	The Random Forest learns from the training data
Evaluate	The model is tested on 500 separate test records

Stage	What happens
Save	The model and the results are stored in the outputs folder
Load	The Streamlit application loads the saved model when it starts
Use	The model helps evaluate suitability for new student profiles

---

17. Advantages of the Project

The project has the following advantages.

17.1 Personalized learning recommendations

Each student gets a list made for them, instead of one common list for everyone.

17.2 Considers individual student preferences

The system takes the student's score, interests, learning style and completed topics into account. This respects the fact that students are different.

17.3 Reduces time spent searching for educational materials

Instead of searching many websites, the student gets five ready suggestions with links. This saves time that can be used for real studying.

17.4 Recommends multiple learning formats

The recommendations can include different resource types, such as videos and practice resources. This gives variety and helps different kinds of learners.

17.5 Uses machine learning for suitability classification

The system does not depend only on fixed rules. A trained Random Forest model learns from data and evaluates suitability.

17.6 Provides an interactive interface

The Streamlit application lets students use the system by entering details and clicking a button. No coding knowledge is required.

17.7 Can be extended for educational platforms

The design can be developed further and connected with real educational platforms in the future.

17.8 Advantages at a glance

No.	Advantage
1	Personalized learning recommendations
2	Considers individual student preferences
3	Reduces time spent searching for educational materials
4	Recommends multiple learning formats
5	Uses machine learning for suitability classification
6	Provides an interactive interface
7	Can be extended for educational platforms

---

18. Limitations and Future Scope

No project is perfect. Being honest about limitations shows maturity and is very good for a viva. The current project has some limitations.

18.1 Limitations

Limitation 1: The training dataset is synthetic. The 2,500 student records were created artificially and not collected from real students. A model trained on synthetic data may not behave the same way on real students, because real learners are more varied and unpredictable.

Limitation 2: Recommendation quality depends on the available learning resources. The system can only recommend from the 104 resources in the dataset. If a useful resource is not in the list, the system cannot suggest it. A better and larger collection would give better recommendations.

Limitation 3: The model has not been validated through large-scale real-student testing. The model was tested on 500 test records, but no large-scale testing with real students has been done. So we cannot say how it will perform in real classrooms.

Limitation 4: The evaluation metrics do not establish actual improvement in academic performance. Accuracy, precision, recall and F1-Score measure the classification model. They do not prove that students who follow the recommendations will score better.

18.2 Limitations summary

Limitation	Effect
Synthetic training dataset	Results may differ with real students
Depends on available resources	Only the 104 resources can be suggested
No large-scale real-student validation	Real-world performance is unknown
Metrics do not prove better academic results	Cannot claim improved learning

18.3 Future scope (future enhancements)

The project can be improved in the following ways.

1. Integrate real student learning records. Using data from real learners would make the model more realistic and trustworthy.

2. Add student feedback to improve recommendations. Students could say whether a recommendation was helpful. This feedback can be used to improve future suggestions.

3. Integrate with online learning platforms. Connecting the system with real platforms would let it work with a much larger library of resources and real users.

4. Track student learning progress. The system could keep a record of what each student has completed over time.

5. Develop a more advanced recommendation engine. More powerful methods could be explored to make recommendations even better.

6. Introduce adaptive recommendations based on previous learning outcomes. The system could change its suggestions as the student learns and improves, so that the next recommendation always matches the current level.

18.4 Future scope at a glance

No.	Future enhancement	Benefit
1	Real student learning records	More realistic model
2	Student feedback	Better recommendations over time

No.	Future enhancement	Benefit
3	Online learning platform integration	Larger resource library and real users
4	Progress tracking	Knowing where each student stands
5	More advanced recommendation engine	Higher quality suggestions
6	Adaptive recommendations	Suggestions that grow with the learner

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19. What Makes This Project Different?

19.1 Not just a website with links

The project is not simply a website containing educational links. It is a machine-learning-based application that takes student information, evaluates learning suitability and generates personalized educational recommendations.

19.2 Three important components

The project combines three important components:

Component	Function
Data Analysis	Understands student learning information
Machine Learning	Evaluates learning suitability
Recommendation System	Suggests relevant educational resources

19.3 Traditional platform versus this project

Point	Traditional platform	This project
Resources shown	Similar for all students	Different for each student
Student profile used	Usually not	Yes: score, interests, style, completed topics
Decision making	Fixed list	Machine learning model evaluates suitability
Results	Many links to search through	Five ranked resources with scores

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Point	Traditional platform	This project
Completed topics	May be shown again	Part of the student profile

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20. Important Project Facts to Remember

This is a quick revision table. Read it before your viva or presentation.

Question	Answer
Project name	Personalized Learning Recommendation System
Developer	Priyasmita Karmakar
Domain	Machine Learning
Main algorithm	Random Forest Classifier
Training records	2,500
Learning resources	104
Testing records	500
Accuracy	86.8%
Precision	94.06%
Recall	90.62%
F1-score	92.31%
Recommendations generated	5
Interface	Streamlit
Language	Python
Saved model file	outputs/suitability_model.joblib
Subject code	BCA57205

20.1 One-paragraph summary to memorize

"My project is the Personalized Learning Recommendation System. It takes a student's academic score, interests, learning style and completed topics, uses a trained Random Forest Classifier to evaluate learning suitability, ranks 104 learning resources, and shows

the five best recommendations with suitability scores in a Streamlit application. The model was trained on 2,500 student interaction records and tested on 500 records, with an accuracy of 86.8%, precision of 94.06%, recall of 90.62% and F1-score of 92.31%."

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21. Viva Preparation Guide

21.1 The most important technical distinction

> **Your model's classification performance and your recommendation engine's suitability scores are two different things. In your viva, do not describe an 87.3% recommendation score as 87.3% model accuracy.**

Say this	Do NOT say this
"The model's accuracy is 86.8% on 500 test records."	"The recommendation accuracy is 86.8%."
"Statistics has a suitability score of 87.3%."	"Statistics is recommended with 87.3% accuracy."
"These are classification metrics."	"This proves students will perform better."
"The training data is synthetic."	"The model was trained on real student data."

21.2 Possible viva questions and simple answers

Q1. What is your project about? It is a system that recommends five suitable learning resources to a student based on their score, interests, learning style and completed topics.

Q2. Which algorithm did you use and why? I used the Random Forest Classifier. It combines many decision trees, handles different types of features, can learn complex relationships, reduces dependence on a single tree, gives a good classification baseline and works well for structured data.

Q3. What is Random Forest? It is a supervised machine learning algorithm that builds many decision trees and combines their outputs to make a prediction.

Q4. What is the difference between supervised and unsupervised learning? In supervised learning the data has correct answers (labels) and the model learns from them. In unsupervised learning there are no labels. Random Forest is supervised.

Q5. What datasets did you use? Two datasets. The Student Learning Interactions dataset with 2,500 records for training and evaluating the model, and the Learning Resources dataset with 104 resources from which recommendations are chosen.

Q6. What columns does the resources dataset have? resource_id, category, topic, resource_type, difficulty, estimated_minutes, description and url.

Q7. How many records were used for testing? 500 test records.

Q8. What are your results? Accuracy 86.8%, precision 94.06%, recall 90.62% and F1-score 92.31%.

Q9. What does precision mean? It tells how many of the positive predictions made by the model were correct.

Q10. What does recall mean? It tells how many of the actual positive cases the model correctly found.

Q11. What is F1-score? It is a combined measure of precision and recall.

Q12. Does 86.8% mean your recommendations are 86.8% accurate? No. 86.8% is the classification accuracy of the model on test records. It is not recommendation accuracy and it is not proof of improved student performance.

Q13. What is a suitability score? It is a percentage given to each recommended resource that shows how well it fits the student. It is used to rank the resources.

Q14. Can you show an example recommendation? For a student interested in Data Science: Statistics (Practice, 87.3%), Data Cleaning (Practice, 85.0%), EDA (Video, 84.5%), Pandas (Video, 84.1%) and NumPy (Practice, 82.3%).

Q15. Why did you use Streamlit? To make an interactive web application so that a student can use the system without running Python scripts manually.

Q16. What is Joblib used for? To save the trained model as `suitability_model.joblib` and load it later without training again.

Q17. What is EDA and which file does it? Exploratory Data Analysis means understanding the data using numbers and charts. The `eda.py` module does it and creates visualizations.

Q18. What is the limitation of your project? The training data is synthetic, quality depends on the available resources, it has not been validated with large-scale real-student testing, and the metrics do not prove improvement in academic performance.

Q19. How can you improve the project? By using real student records, adding student feedback, connecting with online learning platforms, tracking progress, building a more advanced recommendation engine and adding adaptive recommendations.

Q20. What makes your project different from a normal learning website? It is not just a list of links. It combines data analysis, machine learning and a recommendation system to give personalized suggestions.

Q21. What libraries did you use? Pandas, NumPy, Scikit-learn, Joblib, Matplotlib and Streamlit, with Python as the programming language.

Q22. How many recommendations does the system give? Five.

Q23. What inputs does the student give? Academic score from 0 to 100, learning interests, preferred learning style and previously completed topics.

Q24. Why is the test data kept separate from the training data? To check how the model performs on data it has never seen. This gives an honest measure of performance.

Q25. Where are the latest recommendations stored? In `outputs/last_recommendations.csv`.

21.3 Tips for answering in the viva

- Speak in simple words and give a small example.
- Always say "suitability score" for recommendations and "accuracy" only for the model.
- Be honest that the dataset is synthetic. Examiners respect honesty.
- If you do not know something, say what you do know and explain calmly.

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22. Conclusion

The Personalized Learning Recommendation System shows how Machine Learning can be used to make learning easier and more personal. Instead of giving every student the same list of resources, the system studies each student's academic score, interests, learning style and completed topics, and recommends the five most suitable learning resources.

The project uses Python and a Random Forest Classifier trained on 2,500 student learning interaction records, and it chooses recommendations from 104 learning resources. On 500 test records, the model reached an accuracy of 86.8%, a precision of 94.06%, a recall of 90.62% and an F1-score of 92.31%. These are classification metrics and should not be mistaken for recommendation accuracy or for proof that student performance will improve.

The Streamlit interface makes the system easy to use, and the saved Joblib model makes it fast to run. The project is well organised, with separate folders for data, source code, outputs and reports.

The project also has honest limitations: the training data is synthetic, the resource collection is limited, and there has been no large-scale testing with real students. With real student data, feedback, progress tracking and integration with online platforms, it can grow into a more powerful educational tool.

In short, the project demonstrates the combination of **data analysis, machine learning and recommendation** to support personalized learning.

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23. Glossary of Simple Terms

Term	Simple meaning
Machine Learning	Teaching computers to learn patterns from data
Supervised Learning	Learning from data that has correct answers
Classification	Putting things into categories
Decision Tree	A flowchart-like model that decides by asking questions
Random Forest	A model made of many decision trees that combine their answers
Dataset	A collection of data in rows and columns
Record	One row of data
Feature	One piece of information used by the model, like a column
Label	The correct answer attached to a data record
Training	The model learning from data
Testing	Checking the model on separate, unseen data
Synthetic data	Data created artificially, not collected from real people
Accuracy	Share of all predictions that were correct
Precision	Share of positive predictions that were correct
Recall	Share of real positives that were found
F1-Score	A combined score of precision and recall
Confusion Matrix	A table showing correct and wrong predictions

Term	Simple meaning
Recommendation System	A program that suggests items to a user
Suitability Score	A percentage showing how well a resource fits a student
EDA	Exploratory Data Analysis: understanding data with numbers and charts
Streamlit	A tool to build interactive web apps with Python
Pandas	A library to work with tables of data
NumPy	A library for numerical calculations
Scikit-learn	A library for Machine Learning
Joblib	A library to save and load trained models
Matplotlib	A library to draw charts
CSV	A simple file format for storing table data
URL	A web link

Project: Personalized Learning Recommendation System
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